

Investigation of MPI and DGD Tolerance in Digitally CD Pre-Compensated PAM-2 for Very High Speed PON Systems

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Abstract *Digital Chromatic Dispersion Compensation (DCPC) has been investigated through both experiments and simulations to assess its resilience to multipath interference and differential group delay. The results show a performance similar to conventional PAM-2, with MPI-induced penalty below 1 dB for interference values under 15 dB. ©2026 The Author(s)*

Introduction

The recently released ITU-T G.Suppl.88 white paper presents many physical layer solutions for next-generation Very High Speed PON (VHSP) targeting net data rate towards 200 Gb/s. Regarding VHSP solutions based on Direct Detection (DD), chromatic dispersion (CD) impact is clearly one of the main impairments (the others being optoelectronic component bandwidth and the overall power budget). Regarding downstream (DS) transmission, to keep simple DD-based ONUs, one of the solutions proposed in G.Suppl.88 is the introduction of Digital Chromatic Dispersion Pre-Compensation (DCPC) at the transmitter (Tx). In this approach, an amount of CD equal to $D \cdot L$ [ps/nm] is pre-compensated in the Tx DSP by applying a complex valued FIR-filter with an equivalent dispersion $-D \cdot L$. [1–4]

DCPC has been extensively investigated through both simulations and experiments [5, 6]. However, to the best of our knowledge, its resilience to multi-path interference (MPI) and Polarization Mode Dispersion (PMD, or Differential Group Delay DGD) has not yet been addressed. This paper aims to fill this gap and is organized as follows. We first present experimental measurements on the impact of MPI on DCPC-PAM2 transmission. Due to laboratory equipment constraints, the baud rate is set to 50 Gbaud, corresponding to a 50 Gbps line rate. To emulate a challenging scenario in terms of accumulated CD, experiments are conducted in C-band over 20 km.

After validating the accuracy of our Matlab simulation platform against the experimental results, we report the results of an extensive performance study in terms of MPI resilience for a representative VHSP scenario, i.e. 120G baud DCPC-PAM-2 and transmission in the upper O-band. The results show that the impact on the DCPC-PAM-2 performance is comparable to that observed in standard

PAM-2. i.e. an MPI penalty >1 dB starts to become critical when the MPI Signal-to-Interference-Ratio (SIR) is below 15dB.

Finally, we address the impact of PMD. While PMD has been negligible in previous PON standards, this may no longer hold for a 120 Gbaud signals. According to ITU-T G.652, PMD values are below $0.2 \text{ ps}/\sqrt{\text{km}}$ for "new" fibers (with many modern datasheets reporting values below $0.05 \text{ ps}/\sqrt{\text{km}}$), but can reach up to $0.5 \text{ ps}/\sqrt{\text{km}}$ for legacy fibers. The resulting DGD at 20 km and 99.9% percentile is 7.6 ps for legacy fiber, which is critical when compared to the symbol duration $T_s = 8.3 \text{ ps}$ required for 120-Gbaud transmission.

Experimental setup and results

The experimental setup used to evaluate the impact of MPI on DCPC PAM-2 transmission is shown in Fig.1. A 50-Gbaud PAM-2 signal was generated offline and digitally pre-compensated for chromatic dispersion at the OLT transmitter side, by means of FIR filtering. The DSP and DCPC techniques details can be found in [1, 7]. The insets in Fig.1 show the time-domain real-valued signals representing the in-phase (I) and quadrature (Q) components at the output of the DSP pre-compensation block. The I and Q signals were then loaded into a 92-GS/s arbitrary waveform generator (AWG) and applied to the two arms of an IQ Mach–Zehnder modulator (IQ MZM). The optical carrier was provided by a continuous-wave distributed feedback (DFB) laser, and the IQ MZM was biased for effective DCPC operation. An EDFA was then used to set the launch power P_{TX} to 11 dBm.

To emulate MPI, the optical signal was split into a main 20-km SMF path and an interference branch comprising 1-km of SMF, a variable optical attenuator (VOA) to control the signal-to-interference ratio (SIR), and a polarization controller. The two branches were then recombined

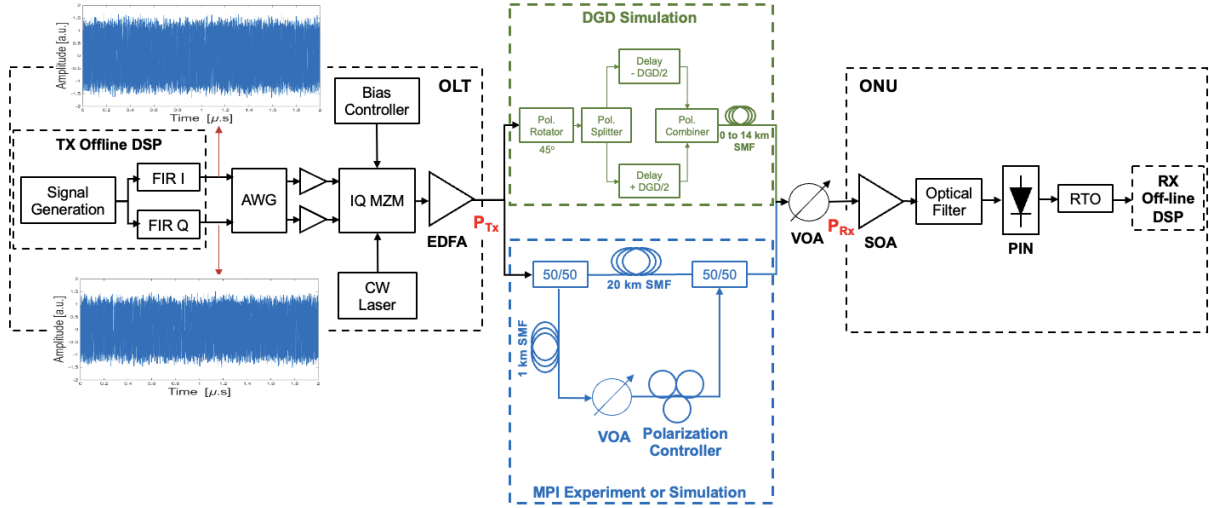


Fig. 1: Block diagram of the MPI experimental setup (blue) including a DGD Block Simulator (green). The insets show the pre-compensated I and Q-component time domain signals at IQ MZM electrical inputs.

at the receiver input, reproducing worst-case interference conditions. An additional VOA was used to emulate the optical distribution network (ODN) loss. At the ONU side, the signal was pre-amplified by a SOA, optically filtered, detected by a broadband PIN photodiode, and acquired by a real-time oscilloscope at 2 samples/symbol for offline DSP. Receiver-side equalization was implemented using an adaptive feed-forward equalizer (FFE) with 100 taps. After PAM-2 hard decision decoding, the BER was evaluated by direct error counting.

Fig. 2a reports the BER as a function of received optical power (ROP) for the full CD compensated case, i.e. $L_{DCPC} = L_{Fiber}$, which corresponds to an amount of pre-compensated CD equal to 340 ps/nm ($=20 \text{ km} \cdot 17 \text{ ps/nm/km}$). At the highest SIR, there is no MPI and the Rx sensitivity at a pre-FEC BER threshold (BER_T) of 10^{-2} is -32 dBm, which corresponds to an ODN loss of 43 dB. The ODN loss is defined as the difference between the transmitted optical power (TOP) measured at the fiber input and the ROP measured at the SOA input. As the SIR decreases, the Rx sensitivity increases as there is more interference.

We summarize the penalty, as function of SIR, measured at $BER_T = 10^{-2}$ and $BER_T = 2 \cdot 10^{-2}$ in Figs. 2b and 2c respectively. The penalty is

defined as the ODN-loss degradation (or. equivalently, the ROP degradation) at BER_T with respect to the corresponding highest-SIR condition. In both plots, markers denote the experimental measurements, while solid lines report the corresponding simulation results. Figs. 2b and 2c also show experimental and simulation results obtained in two other DCPC conditions: 75% compensation ($L_{DCPC} = 3/4 \cdot L_{Fiber}$), and half compensation ($L_{DCPC} = 1/2 \cdot L_{Fiber}$). As a reference, Figs. 2b and 2c also include a benchmark curve (solid black line) reporting the MPI tolerance of a standard PAM-2 signal (without DCPC) affected by negligible CD, showing a similar behavior to the fully pre-compensated case. The experimental results appear slightly better than simulations likely due to polarization variations that avoid the worst-case conditions.

In both figures, full CD compensation provides the highest robustness to MPI and even at SIR of 10-dB, the penalty is limited to ~ 2 dB indicating strong resilience under severe interference conditions. In contrast, partial compensation exhibits a greater sensitivity to SIR reduction, suggesting that a mismatch between the applied CDPC and the actual CD reduces the interference tolerance. This is reflected in the additional penalty observed with

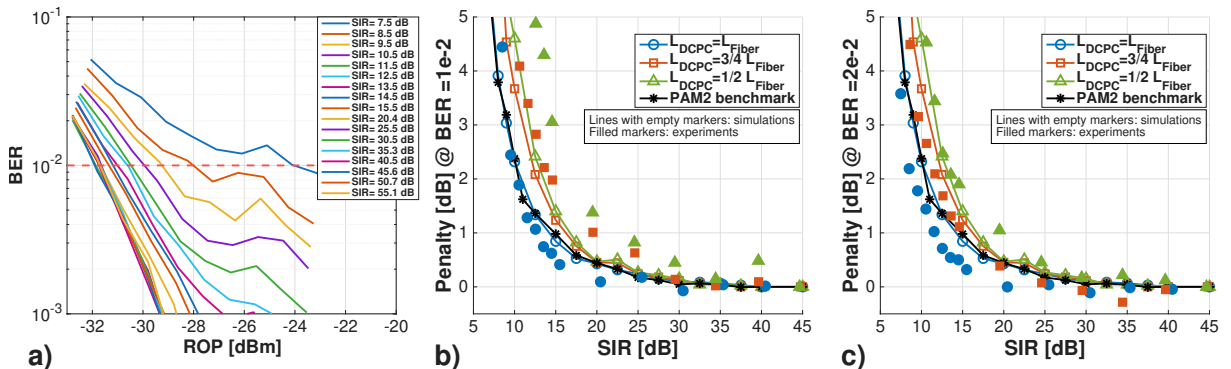


Fig. 2: a) Experimental BER versus ROP @ $L_{DCPC} = L_{Fiber}$ for different SIR values. b) ODN loss penalty vs. SIR at $BER_T = 10^{-2}$. c) Same as (b) evaluated at $BER_T = 2 \cdot 10^{-2}$ (Lines with empty markers: simulations, filled markers: experiments). As a reference, the penalty for a standard PAM-2 signal (without DCPC) is shown (black line).

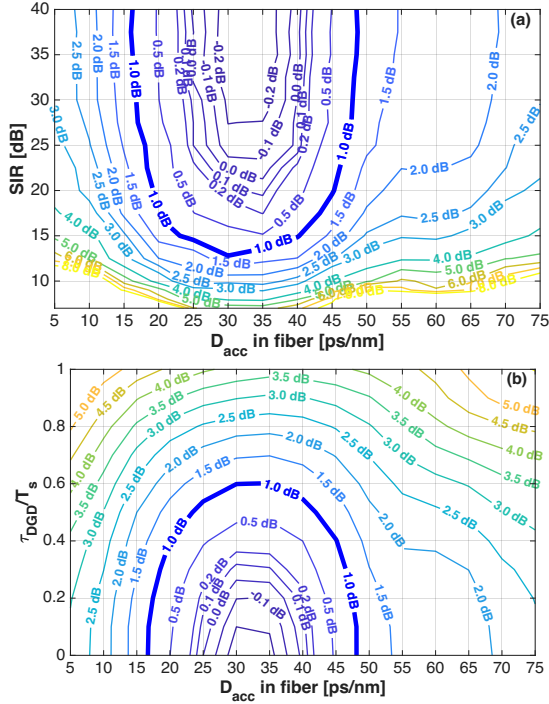


Fig. 3: (a) Contour plot of the ODN loss penalty as a function of the dispersion accumulated in fiber propagation and the SIR. (b) Contour plot of the ODN loss penalty as a function of the dispersion accumulated in fiber propagation and the normalized DGD. In both plots, the target BER is $2 \cdot 10^{-2}$ and the amount of pre-compensated CD is 30 ps/nm.

partial compensation compared to full CD compensation. These results indicate that a maximum CD tolerance of DCPC must be carefully considered to avoid significant penalties due to MPI.

Scalability study to 120G/λ in O-band using VHSP expected requirements

To assess the scalability of the proposed DCPC PAM-2 solution toward next-generation VHSP systems, we performed a comprehensive simulation study targeting 120G/λ operation in the O-band. First, the simulation framework was validated against the experimental results obtained at 50 Gbaud in the C-band. The results are shown in Figs. 2b and 2c, indicating a good agreement in predicting the performance impact of MPI, with an estimation error within the experimental measurement uncertainties.

Based on this validation, the study was extended to the 120-Gbaud O-band scenario. The simulation set-up is the same already detailed in Fig. 1, assuming a transmitter and receiver electro-optical bandwidth of 65 GHz. The fiber dispersion and attenuation were 5.96 ps/nm/km and 0.35 dB/km, respectively. Fiber nonlinearity had a negligible impact. In the 120 Gbaud simulations, a 5-tap DFE was added after the FFE equalizer to improve performance in the presence of residual CD.

The impact of MPI was evaluated by varying the SIR and measuring the resulting penalty in terms of ROP or equivalent ODN loss at the target

BER of $2 \cdot 10^{-2}$. The results in Fig. 3a, obtained with a CD pre-compensation value of 30 ps/nm, show that DCPC maintains a strong resilience to MPI even at higher symbol rates, with full CD pre-compensation providing the best performance across all considered conditions. Compared to the 50-Gbaud case, the increased symbol rate leads to a higher sensitivity to MPI due to the reduced symbol duration, making interference-induced distortions more critical. Nevertheless, the overall penalty trends remain consistent, confirming the scalability of DCPC in terms of MPI tolerance.

In addition, the robustness of the system to polarization-related impairments was investigated by introducing differential group delay (DGD) in the simulations. A worst-case scenario was considered by splitting the signal at the fiber output into two orthogonal polarization components with equal power and introducing a variable time delay τ_{DGD} between them (see the green block scheme in Fig. 1). The DGD normalized to the symbol duration was jointly varied together with the accumulated CD along the fiber. The resulting penalty map obtained for a CD pre-compensation of 30 ps/nm (corresponding to ~ 5 km) is shown in Fig. 3b. The results indicate that the DCPC PAM-2 solution maintains acceptable performance even in the presence of significant DGD, particularly when the applied pre-compensation is properly matched to the transmission distance. Specifically, the contour plot in Fig. 3b shows that penalties remain below 1 dB for normalized DGD values up to $0.5T_s$ (~ 4.1 ps) whenever the difference between the actual value of the accumulated dispersion in the fiber and the DC pre-compensation value is less than $\sim 20\%$. This indicates that typical PMD values expected in modern fibers may introduce some penalty but do not represent a limiting factor. However, in worst-case legacy fiber scenarios, where DGD approaches the symbol duration, a significant penalty arises, highlighting the need for accurate DCPC tuning.

Conclusions

We investigated the impact of MPI and DGD on DCPC PAM-2 transmission for VHSP systems through experiments and simulations. Results at 50 Gbaud show that full pre-compensation ensures strong robustness to MPI, while partial compensation may lead to additional penalties. Simulations at 120 Gbaud confirm that MPI tolerance remains comparable, with penalties becoming significant for SIR below ~ 15 dB. DGD is shown to be non-limiting for typical modern fibers, but can become critical in worst-case legacy scenarios when approaching the symbol duration. Overall, DCPC PAM-2 appears as a viable solution for 100G/λ for VHSP systems, provided that accurate dispersion matching is ensured.

Acknowledgements

This work was carried out in the PhotoNext Center facilities at Politecnico di Torino (www.photonext.polito.it).

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